

Seat. No. _____

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BE SEMESTER-8 (Computer Engineering) Examination April - 2025

Subject Name: Web Data Management

Subject Code: CE803D-N

Date: 11/04/25

Time: 12.30 to 3.30

Total Marks: 70

Instructions:

1. Answer each section in separate answer sheet.
2. Use of scientific calculator is permitted.
3. All questions are Compulsory.
4. Indicate clearly, the option you attempt along with its respective question number.
5. Use the last page of main supplementary of rough work.

Section-I

Q-1 (A) What are the key differences between structured, semi-structured, and unstructured data? Explain with examples. [5]

(B) How does XML Schema (XSD) help in defining constraints on XML documents? [5]

(C) What is XPath? Explain its axes and node tests with examples. [5]

OR

(C) Explain Holistic Twig Joins in XML query processing and why they are efficient. [5]

Q-2 (A) How do region-based identifiers help in XML document retrieval? [5]

(B) How do updates affect XML identifiers, and what techniques can be used to maintain identifier consistency? [5]

OR

Q-2 (A) How does XPath 2.0 improve upon XPath 1.0? List key enhancements. [5]

(B) What are the challenges in querying RDF data? How can they be addressed? [5]

Q-3 (A) What is DL-LITE ontology and how is it used for query answering? [5]

(B) Explain web graph mining with an example. [5]

OR

Q-3 (A) How does inference in RDF and OWL improve data retrieval? [5]

(B) Explain the concept of failure management in distributed systems. [5]

Section-II

- Q-4 (A) Explain XML Data Model and how it differs from the relational data model. [5]
(B) Describe the role of XLink and XPointer in XML. How do they help in linking XML documents? [5]
(C) Explain the FLWOR expressions in XQuery with an example query. [5]

OR

- (C) Describe Dewey-based identifiers in XML. How do they help in XML indexing? [5]

- Q-5 (A) Explain Fragmentation of XML document in detail. [5]
(B) What is the difference between absolute and relative XPath expressions? Provide examples [5]

OR

- Q-5 (A) Discuss different optimization techniques for XML query evaluation. [5]
(B) Explain the concept of Peer-to-Peer (P2P) Data Management Systems. [5]

- Q-6 (A) What are the advantages and challenges of Ontology-Based Data Integration (OBDI)? [5]
(B) Describe the MapReduce programming model with an example. [5]

OR

- Q-6 (A) How does peer-to-peer (P2P) networking enhance data sharing? [5]
(B) What is PageRank? How is it used in ranking web pages? [5]
